

ML / AI Tier

The ML tier (Tier 3) provides **advisory** anomaly detection, distribution drift monitoring, and a provenance-tracked model registry. It is implemented in `src/ml/` and surfaced under `/api/v1/ml`.

The advisory-only guarantee

The ML/AI tier can inform operators. It can **never** command a mission-state change or alter Tier-1/Tier-2 data.

This is enforced structurally, not just by convention:

- `MLAnalysisWorker` holds **no reference** to the mission state machine.
- No file under `src/ml/` imports or references `MissionStateMachine` / `mission_machine`. This is asserted by `tests/test_ml.py::test_ml_source_does_not_reference_mission_state_machine`.
- Every ML API response carries an advisory note and cannot mutate mission state.

Detectors

Detector	Dependency	Availability	Method
<code>robust_zscore</code>	none (pure NumPy)	Always available	Median/MAD robust z-score: $ x - \text{median} / (1.4826 \cdot \text{MAD})$, cutoff 3.5. Zero-MAD features are guarded, not turned into infinities.
<code>isolation_forest</code>	scikit-learn	If sklearn imports	Isolation Forest, contamination='auto', 200 estimators.
<code>gmm</code>	scikit-learn	If sklearn imports	Gaussian Mixture negative log-

Detector	Dependency	Availability	Method
			likelihood, 5th-percentile cutoff.
lstm_forecast	PyTorch	If torch imports	One-step LSTM forecast; residual z-scored, cutoff 3.0.
ensemble	—	Uses whichever of the above are available	Rank-normalises each available detector's scores, averages them, and flags a sample when a majority ($\geq 50\%$) of contributors vote anomalous. Records exactly which detectors contributed.

GET /api/v1/ml/detectors returns a **live** availability probe: for each detector it reports available, the method, and — when unavailable — the exact reason (e.g. the import error). Nothing is claimed to work that cannot.

Feature extraction — no imputation

extract_feature_matrix builds the numeric matrix from a window of telemetry frames. A frame contributes a row **only if every requested feature is present and valid** in that frame. Incomplete frames are dropped and counted (dropped_incomplete); they are never zero-filled or imputed. The default feature set is:

bme688_temperature, bme688_pressure, bme688_humidity,
gnss_altitude, radiation_cpm, battery_voltage

Drift monitoring

population_stability_index(expected, actual) computes the PSI between a reference distribution and a current window:

PSI	Classification
< 0.10	STABLE
< 0.25	MODERATE
≥ 0.25	SIGNIFICANT
insufficient data	UNAVAILABLE (returns None, never a fabricated 0)

The DriftMonitor stores a reference matrix and evaluates per-feature PSI on demand. Insufficient data yields an explicit unavailable result.

Training

POST /api/v1/ml/train (SCIENTIST role, rate-limited 2/hour) runs train_detector over stored telemetry:

1. Reconstruct frames from persisted measurements.
2. Extract the feature matrix (dropping incomplete frames honestly).
3. Refuse to train if fewer than **20** valid samples — returns 422 with the exact count and how many frames were dropped. **No fake model is created.**
4. Fit and score the detector.
5. Persist the artefact via ModelStore and record its provenance.

Model provenance & integrity

Every trained model records:

- training_data_hash — a hash of the exact feature matrix it was trained on.
- sha256_hash — the SHA-256 of the serialised artefact on disk.

ModelStore.load(path, expected_sha256) **recomputes** the file's SHA-256 and refuses to load if it is missing, if no expected hash is supplied, or if the hash does not match. A tampered model file therefore fails closed with a SafeLoadError rather than executing an untrusted pickle. This is verified by tests/test_ml.py::test_model_store_detects_tampering.

Live advisory summary

The MLAnalysisWorker maintains a rolling window and periodically scores it with the ensemble. GET /api/v1/ml/anomalies returns the latest summary with an honest state:

State	Meaning
IDLE	Worker running, no analysis yet.
INSUFFICIENT_DATA	Fewer than the minimum samples in the window.
UNAVAILABLE	No detector could run.
AVAILABLE	A real anomaly summary is present.
NOT_CONFIGURED (HTTP 503)	The ML worker is not running at all.

At no point does the tier invent a “clean” result to appear healthy.